

Constructing the Higman-Sims Graph in Orbiter

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1 Introduction

Constructing interesting combinatorial objects can be somewhat intriguing for a beginner. A combinatorial object comes with an automorphism group, and many interesting objects have large groups. This raises the chicken and egg problem: which was first, the group or the object? Take for example the Golay code. The code is a subspace of dimension 12 of a 24-dimensional vectors space over the binary field with minimum distance 8. The automorphism group is the Mathieu group M_{24} . Given the group, it is easy to construct the code. Given the code, it is possible to construct the group. But this is impractical for a beginner. Where do we start? Should we first learn about the code or should we start with the group? What if we want to start from scratch? We need to have a step-by-step learning process that does not require substantial previous knowledge. This is a big hurdle in making combinatorics accessible to a wider audience. The present paper is contribution to making this step-by-step process possible in one case. Namely we focus on an interesting strongly regular graph known as the Higman-Sims graph. The construction of this graph is well-known, but the steps are hard to do. Computer algebra is used in a meaningful way to perform the construction step-by-step. The result will be an interesting combinatorial object, detailed information about the automorphism group, and information about properties. The construction itself is supported by the computer algebra system Orbiter, but the steps are visible for the user. We believe that this is helpful to make an interesting topic in the theory of algebraic combinatorics accessible to a wider audience. This will teach the learner about the beautiful connections between group theory, combinatorics, geometry, coding theory and graph theory.

A better way to motivate the object is by starting from a general set of conditions that we wish to satisfy. For the code, we may ask that the code corresponds to a projective set of size 24 in the projective geometry $PG(11, 2)$, and that any 7 points are independent. A tedious but manageable computation using the poset classification algorithm shows that the set is unique and that it has a stabilizer of order 244823040. The group is of course the Mathieu group M_{24} though the computer does not know this.

Once we have the set of points, we can fill the coordinate vectors in the columns of a matrix to create a check matrix of the extended Golay code, from which a generator matrix can be computed. It turns out that the code is self-dual, so the check matrix is also a generator matrix. By construction, the code has minimum distance at least 8 (any 7 columns of the check matrix are independent). It cannot be more than 8, so it is exactly 8.

2 Preliminaries

Before we can get to work, we need to get accustomed to some conventions with how Orbiter works. One important idea is that of indexing. This is a way of encoding objects using integers from an interval, typically starting with either 0 or 1. During indexing, it does not really matter how the bijection between objects and integers is chosen. However, it matters that the bijection is fixed. So, the ranks of objects are permanent. Simply put, indexing is a way of reducing the memory footprint of combinatorial data, as the index values are often more compact than the descriptions of the objects. The indexing function is deterministic: the same object always gives the same index value. Orbiter provides many different index functions. Orbiter index values are zero-based, so the lowest index value is 0 (and not 1).

Let us consider the projective space. We denote by $\text{PG}(n, q)$ the n -dimensional projective space over the field $\mathbb{F}_q$. This is an incidence geometry with finitely many objects called points. The points are structured by subspaces. The basis is a vector space. Let $V(d, q)$ be the d -dimensional vector space over $\mathbb{F}_q$. The projective space $\text{PG}(n, q)$ arises from $V(n+1, q)$ in the following way. Let $\mathfrak{Gr}_{d,k,q}$ be the Grassmannion of k -subspaces of $V(d, q)$. The points of $\text{PG}(n, q)$ are the elements of $\mathfrak{Gr}_{n+1,1,q}$, i.e., the one dimensional subspaces. The lines are the elements of $\mathfrak{Gr}_{n+1,2,q}$. Lines are viewed as sets of point. More generally, we can consider projective subspaces of i as $i+1$ subspaces of $V(n+1, q)$. again, it is customary to think of these subspaces as sets of points, simply by inclusion. There are formulae for the number of subspaces of any given dimension. Using the Gauss coefficients

$$\begin{bmatrix} d \\ k \end{bmatrix}_q = \prod_{i=0}^{k-1} \frac{q^d - q^i}{q^k - q^i},$$

we find that

$$|\mathfrak{Gr}_{d,k,q}| = \begin{bmatrix} d \\ k \end{bmatrix}_q.$$

In particular, the number of points of $\text{PG}(n, q)$ is

$$\begin{bmatrix} n+1 \\ 1 \end{bmatrix}_q = \frac{q^{n+1} - 1}{q - 1}.$$